Minimum-Cost Temporal Walk Profiles on Depth-Three In-Arborescences

Exact optimization parameterized by preservation loss

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Abstract

This paper studies minimum-cost deletion of explicitly listed temporal events subject to preserving a prescribed number of two-event walks ending at a designated root. For temporal in-arborescences of depth at most three, the algorithm computes the minimum deletion cost for every attainable exact pair of surviving two- and three-event walk counts. Events have nonnegative binary-encoded costs and may be required to survive or to be deleted. If R is the allowed loss of two-event walks after mandatory deletions, the algorithm runs in $2^{O(\sqrt{R})}\,\mathrm{poly}(L)$ time and polynomial space, where L is the complete encoded input length. The unit-cost formulation admits a polynomial-size generalized deletion propagation encoding, and binary deletion costs admit an explicit weighted ILP extension. The result concerns the specialized algorithm and its dependence on preservation loss. The argument charges deleted final events to preservation loss, groups them by continuation-equivalent neighborhoods, and enumerates a partition-bounded family of branch configurations. Chronological dynamic programs recover exact profiles and original-event witnesses. For contrast, the corresponding unit-cost quota decision problem is NP-complete and W[1]-hard in the deletion budget on the same graph class. The profiles support arbitrary efficiently evaluated predicates of the two counts and deletion-invariant data; an elementary hardness example delineates an additional dynamic-feature boundary. An exhaustive check on small instances compares the reference implementation with direct enumeration. A study of synthetic transaction histories illustrates why a three-walk upper bound and the corresponding exact target can require different answers.

1 Introduction

Deleting a temporal event can destroy several time-respecting interactions at once. A decision may therefore have two competing effects: it can suppress longer interactions while also removing shorter ones that are required to survive. The model captures this tradeoff on a directed in-arborescence of depth at most three, with all walks ending at its root. Parallel contacts are separate events, and successive timestamps must increase strictly. The algorithm chooses a shared set of event deletions, preserving a lower quota on two-event walks and recording exactly which two- and three-event walk counts can remain.

Temporal reachability modification and path counting already have substantial algorithmic foundations. Molter, Renken, and Zschoche [17, Theorems 14–15] give polynomial forest algorithms for multiple-source temporal reachability minimization by deletion or delay, and Enright, Meeks, and Molter give a polynomial forest algorithm for counting temporal paths [7]. Generalized deletion propagation (GDP) also provides a framework for simultaneous deletion and preservation requirements over query outputs [13]. Appendix E gives a polynomial-size encoding of the present unit-cost formulation in GDP and an explicit weighted ILP extension for binary deletion costs.

The contribution here is the specialized temporal algorithm and its preservation-loss bound. Its sufficient state preserves the multiplicities needed for exact two- and three-event counts while optimizing one shared set of original-event deletions. The complete profile can also be obtained through polynomially many exact target optimizations; producing that table alone is therefore not the algorithmic distinction. Section 9 separates these representation issues from the parameter dependence required to derive Theorem 2.1.

The preservation parameter. After applying mandatory deletions, let N_2 be the baseline number of two-event walks and let T_0 be the nonnegative preservation quota. The allowed additional loss is $R = N_2 - T_0$. The algorithm computes every admissible exact count pair, its minimum nonnegative binary-encoded deletion cost, and an attaining witness in $2^{O(\sqrt{R})}\,\mathrm{poly}(L)$ time and polynomial space (Theorem 2.1). The polynomial exponent is absolute. Neither a monetary value nor the number of branches enters the exponential factor. The input is an explicit event list; binary multiplicities standing for unlisted events are outside the model.

Where the bound comes from. A final event is a contact incident to the root. Its deletion destroys a number of baseline two-event walks equal to the number of earlier middle events. These charges are disjoint across final events. Within one branch, equal charges identify equal prefix neighborhoods and therefore equal effects under every continuation on the other event layers. Nonnegative costs allow a canonical cheapest choice within each class. The possible charged multiplicities are bounded by the number of integer partitions of numbers at most R. The algorithm enumerates these configurations separately for each branch. For a fixed configuration, a chronological state records the retained first-event prefix count, exact two-event loss, and exact three-event count. Exact convolutions then assemble child and branch profiles. This order of operations keeps one partition factor rather than a product over branches.

Why exact profiles matter. A scalar optimum for a monotone quota can discard intermediate count values that are essential for a nonmonotone target. Moreover, two unfinished choices can have the same current cost and walk counts yet admit different optimal continuations. Proposition 3.4 gives a ten-event example where keeping either one of two scalar-equal contexts loses a profile entry or its minimum cost. The state keeps the information needed until it is safe to merge completed tables. Retaining exact coordinates, minimizing values, and min-plus convolution are standard techniques; the structural contribution is their sufficient temporal state and the preservation-loss bound proved here.

Complexity boundaries and supported decisions. The unit-cost quota problem is W[1]-hard in the deletion budget even when every event is optional (Theorem 6.1); its classical decision version is NP-complete (Corollary A.7). This is a contrast between parameters on the same graph class, not a lower bound in R. Once a full profile is available, any efficiently evaluated acceptance predicate of the two counts and invariant auxiliary data can be optimized by a scan. This includes nonmonotone count-based classifiers. An arbitrary additional dynamic amount feature already permits NP-completeness at $R = 0$ (Proposition 7.2), so exactness does not extend automatically to a richer feature pipeline.

The remaining sections give the full structural proof, the companion hardness result, concrete profile examples, related-work comparisons, and validation. The complete budget-hardness construction and a counting-model normalization are included in the appendices. The implementation evidence concerns correctness on a finite corpus; no practical speed or application-effectiveness claim is needed for the formal result.

2 Model and main result

2.1 Events, counts, and normalization

Let E be a set of M explicitly listed events, each with a unique original identifier, a source vertex, a target vertex, an integer timestamp $t(e)$, and a nonnegative integer deletion cost c_e. Costs and timestamps are encoded in binary for the complexity statement. Parallel events are distinct decisions, including events with identical endpoints and timestamps. After parallel arcs are collapsed, the original graph is an in-arborescence directed toward a designated root z, with depth at most three. Every mentioned vertex has a path to z. The empty history is allowed. Depth and roles refer to this original graph, even if subsequent mandatory deletions disconnect it.

For $S \subseteq E$ and $h \in \{2, 3\}$, define $W_h(S)$ as the number of ordered h-tuples of events of S that form a directed walk *ending at* z, with strictly increasing timestamps. Walks ending elsewhere are excluded. In particular, a path of three increasing events has $W_2 = 1$ and $W_3 = 1$, although it has two length-two subwalks if endpoints are unrestricted.

Let T be an integer preservation quota, and let F and K be disjoint forced-deleted and forced-retained sets. All other events are optional. Set

$$E_0 = E \setminus F, \qquad O = E_0 \setminus K, \qquad N_2 = W_2(E_0), \quad N_3 = W_3(E_0), \quad T_0 = \max\{T, 0\}.$$

If $T_0 > N_2$, no deletion can meet the lower quota and the instance is rejected. Otherwise let

$$R = N_2 - T_0, \qquad \lambda(D_0) = N_2 - W_2(E_0 \setminus D_0), \quad D_0 \subseteq O. \tag{1}$$

Then $0 \le R \le M^2$, and the quota is exactly $\lambda(D_0) \le R$. Forced cost $c(F) = \sum_{e \in F} c_e$ is reserved once. It is never charged again inside a dynamic-programming transition. The input length L includes the complete explicit encodings of identifiers, graph, timestamps, statuses, costs, and quota.

2.2 Exact minimum-cost profiles

For $0 \le \ell \le R$ and $0 \le w \le N_3$, define

$$C(\ell, w) = \min \left\{ \sum_{e \in D_0} c_e : D_0 \subseteq O, \quad W_2(E_0 \setminus D_0) = N_2 - \ell, \quad W_3(E_0 \setminus D_0) = w \right\}, \tag{2}$$

with value ∞ if the set is empty. A finite entry must include an attaining set of original identifiers. Its total cost is $c(F) + C(\ell, w)$ and its full deletion witness is $F \cup D_0$.

Theorem 2.1 (Weighted exact attainable profiles)**.** *For the model above, all finite entries of (2), their minimum costs, and attaining original-event witnesses can be computed in*

$$2^{O(\sqrt{R})}(L+1)^{O(1)}$$

time and polynomial space. The polynomial exponent does not depend on R, the number of branches, or the numerical magnitudes of the deletion costs. Strict timestamp ties, parallel events, and both forced statuses are supported. The output contains at most $(R+1)(N_3+1)$ entries.

The output is complete over the *count pairs*, with one minimum-cost witness per pair. It does not list every attainable monetary cost or every witness. With unit costs, selecting the cheapest entry satisfying $w \le U$ solves the corresponding upper/lower quota problem. A total monetary budget can be imposed by filtering entries. An additional, independent cardinality cap in a weighted instance is not represented by this table; it would require additional state.

3 Final-event structure and bounded enumeration

For each depth-one vertex u, let C_u be the events $u \to z$ in E_0. For each depth-two child v of u, let B_v be the events $v \to u$ in E_0, and let A_v contain all events in E_0 from depth-three children into v. Put $B_u = \bigcup_{v \to u} B_v$. These event domains are disjoint. Original children remain present in the decomposition even if their surviving event lists are empty. For a retained $b \in B_v$, let

$$p_b = |\{a \in A_v \setminus D_0 : t(a) < t(b)\}|, \qquad s_b = |\{c \in C_u \setminus D_0 : t(b) < t(c)\}|.$$

The unique middle event identifies each counted walk, so

$$W_2(E_0 \setminus D_0) = \sum_u \sum_{v \to u} \sum_{b \in B_v \setminus D_0} s_b, \qquad W_3(E_0 \setminus D_0) = \sum_u \sum_{v \to u} \sum_{b \in B_v \setminus D_0} p_b s_b. \tag{3}$$

Deleting an A event cannot affect W_2 under the root-ending definition.

Lemma 3.1 (Final-event charge and equal-degree equivalence)**.** *For $c \in C_u$, write*

$$N(c) = \{b \in B_u : t(b) < t(c)\}, \qquad d(c) = |N(c)|.$$

Every optional deletion set satisfies

$$\sum_u \sum_{c \in D_0 \cap C_u} d(c) \le \lambda(D_0). \tag{4}$$

Within one branch, C events of equal degree have identical neighborhoods. For any fixed decisions on all A and B events, the two walk counts depend only on the number of deleted optional C events of each degree, not on their identities.

Proof. Deleting c destroys the $d(c)$ baseline pairs (b, c). Pairs charged to different final events are disjoint, including when some middle events are also deleted. They are a subset of all destroyed baseline pairs, proving the inequality. Sort B_u by timestamp and break ties in any fixed way. Each $N(c)$ is a prefix of that same order. Strict comparison excludes a whole timestamp tie; it never selects only part of one. Equal-size prefixes are therefore identical. Under fixed A and B decisions, each retained final event of the same degree has the same retained middle neighbors and the same number of preceding retained A events through each of those neighbors. Its pair and triple contributions are identical. □

Corollary 3.2 (Minimum-cost canonical final choices)**.** *For each branch and each positive degree $j \le R$, order the optional C events by nondecreasing cost, breaking cost ties by any fixed identifier order. For each multiplicity x_j, it suffices to delete the cheapest x_j identities in that group. A feasible deletion never removes a C event of degree $j > R$. All optional degree-zero C events may be retained when computing minimum-cost count profiles.*

Proof. The exchange in Lemma 3.1 preserves both counts for every continuation. Replacing a chosen size-x_j subset by the cheapest such subset can only decrease cost and never uses a forced-retained identity. The charge inequality excludes degrees above R. A degree-zero final event participates in no baseline pair and hence no triple; restoring it changes neither count and cannot increase cost because deletion costs are nonnegative. □

If $m_{u,j}$ optional events of degree j exist in branch u, enumerate

$$\mathcal{X}_u = \left\{ (x_j) : 0 \le x_j \le m_{u,j}, \quad \sum_{j=1}^{R} j x_j \le R \right\}. \tag{5}$$

Only present positive-degree groups need explicit coordinates. Groups from different branches are not merged. Configurations are processed one branch at a time; no Cartesian product of all branches' choices is enumerated.

Lemma 3.3 (Configuration bound)**.** *Let*

$$P_{\le R} = \left|\{(x_1,\dots,x_R) \in \mathbb{Z}_{\ge 0}^R : \textstyle\sum_j jx_j \le R\}\right|, \qquad P_{\le 0} = 1.$$

For $R > 0$, $|\mathcal{X}_u| \le P_{\le R} \le \exp(\pi\sqrt{2R/3})$. If g_u positive-degree groups are present, a depth-first enumeration visits at most $(g_u+1)P_{\le R}$ nodes.

Proof. The unrestricted multiplicity vectors are integer partitions of integers between zero and R. Capacities remove vectors. For completeness, the partition generating function gives, for every $\alpha > 0$,

$$P_{\le R} \le e^{\alpha R}\prod_{j\ge 1}(1-e^{-\alpha j})^{-1}, \qquad \log\prod_{j\ge 1}(1-e^{-\alpha j})^{-1} = \sum_{h\ge 1}\frac{1}{h(e^{\alpha h}-1)} \le \frac{\pi^2}{6\alpha}.$$

The last inequality uses $e^x - 1 \ge x$ and $\sum_{h\ge 1} h^{-2} = \pi^2/6$. Taking $\alpha = \pi/\sqrt{6R}$ proves the stated bound, an elementary form of the classical partition estimate [9]. Enumerate each present group's count from zero to the minimum of its capacity and the remaining loss allowance divided by its degree. Every partial vector has a feasible zero extension. At a fixed depth these extensions are distinct, so there are at most $P_{\le R}$ nodes per depth. Counting the $g_u + 1$ depths proves the traversal bound. □

For a fixed $x \in \mathcal{X}_u$, let $D_C(x)$ be the canonical final-event deletions, $\Lambda_x = \sum_j jx_j$, and $\kappa_x = \sum_{c\in D_C(x)} c_c$. For $b \in B_u$, set

$$s_b^x = |\{c \in C_u \setminus D_C(x) : t(b) < t(c)\}|.$$

The two-walk loss inside this branch is exactly

$$\Lambda_x + \sum_{b\in D_0\cap B_u} s_b^x. \tag{6}$$

This is disjoint accounting: first remove pairs ending at deleted C events; then remove pairs with deleted B events and retained C events. It remains correct when both events of a baseline pair are deleted.

3.1 Why unfinished configurations cannot be merged by scalar counts

Proposition 3.4 (Equal current statistics, different exact continuations)**.** *There is a ten-event instance with two final-event deletion configurations that have the same deletion cost, two-walk loss, and current three-walk count, but for which retaining only either one as a representative prevents recovery of the complete minimum-cost profile.*

Proof. Use one root child u and four children $v_1,\dots,v_4$ of u. Retain $B_i : v_i \to u$ at time $2i$, for $1 \le i \le 4$. Make $C_j : u \to z$ at time $2j+1$ optional, for $1 \le j \le 4$. There are two optional first events, $A_2 : s_2 \to v_2$ and $A_4 : s_4 \to v_4$, both at time one. All costs are one. The baseline counts are $(N_2, N_3) = (10, 4)$. Set $T_0 = 5$, so $R = 5$.

The final-event degrees are $1, 2, 3, 4$. Consider $D_{14} = \{C_1, C_4\}$ and $D_{23} = \{C_2, C_3\}$, before either first event has been deleted. Both cost two, lose five two-walks, and leave two three-walks. Their suffix-count vectors on $B_1,\dots,B_4$ are respectively $(2,2,1,0)$ and $(2,1,1,1)$. Completing the optional A decisions gives:

Final-event context	Cost at $W_3 = 2$	Cost at $W_3 = 1$	Cost at $W_3 = 0$
D_{14}	2	∞	3
D_{23}	2	3	4
Pointwise minimum	2	3	3

Costs include final-event deletions. Since all middle events must survive, exact two-walk loss five can arise only from the degree subsets $1 + 4$ and $2 + 3$. First-event deletions change no two-walk count. Keeping only D_{14} loses the attainable coordinate $W_3 = 1$; keeping only D_{23} gives an incorrect minimum cost at $W_3 = 0$. These are the only two relevant final contexts at that exact loss, proving the claim. □

This example rules out the stated one-representative compression. It is not a lower bound on all possible state representations or algorithms. In the algorithm below, each final-event configuration remains fixed until its child tables are complete; pointwise minimization is then safe.

4 An exact chronological dynamic program

Fix one branch configuration x and a child v. Sort $A_v \cup B_v$ by time, processing *all B events before all A events at the same timestamp*. Let $J_i(q, \ell, w)$ be the minimum optional deletion cost among decisions on the first i events such that:

- q is the number of retained A_v events processed so far;
- ℓ is the sum of s_b^x over processed deleted B_v events;
- w is the number of triples contributed by processed retained B_v events, using retained $C_u \setminus D_C(x)$ events.

An unattainable state has value ∞. Initialize only $J_0(0, 0, 0) = 0$. Every transition writes to a fresh layer. The four possible transitions are listed below; a deletion is allowed only for an optional event.

Event	Decision	New state	Added cost
$a \in A_v$	retain	$(q + 1, \ell, w)$	0
$a \in A_v$	delete	(q, ℓ, w)	c_a
$b \in B_v$	retain	$(q, \ell, w + qs_b^x)$	0
$b \in B_v$	delete	$(q, \ell + s_b^x, w)$	c_b

Reject a transition with $\ell > R - \Lambda_x$. All w values remain distinct. There is no rule that discards a larger w because a smaller value was found at the same or a lower cost. The final child table is

$$Q_{v,x}(\ell, w) = \min_q J_{|A_v|+|B_v|}(q, \ell, w).$$

Each finite state stores one attaining original-ID deletion mask alongside its cost. A keep transition preserves the mask; a deletion adds that event's bit; a lower-cost replacement replaces both value and witness.

Lemma 4.1 (Exactness of the child table)**.** *For fixed x, the recurrence computes the minimum optional $A_v \cup B_v$ deletion cost for every attainable exact pair (ℓ, w) with $0 \le \ell \le R - \Lambda_x$, with a valid witness.*

Proof. Use induction on the chronological position. For an A event, retention adds one to the retained prefix count and deletion adds its cost. It affects no root-ending two-walk. For a B event, the tie rule ensures that exactly q retained A events have strictly earlier timestamps. Retaining it contributes

exactly qs_b^x triples. Deleting it destroys precisely its s_b^x remaining pairs, as in (6). Forced-retained events have only the keep transition. These cases cover all legal decisions and each has the stated effect.

At a fixed processing position, histories with the same (q, ℓ, w) have identical possible future decisions and effects. Every previously processed retained A event precedes every future eligible B event. Within a shared timestamp, no A event has been processed before its tied B events, so this claim also covers ties. Only the number of past retained A events is needed for a future prefix; past B contributions have already been included in w. Future event costs and forced statuses are fixed by the input and position. Consequently, keeping only the minimum cost at an identical state is sound. All loss increments are nonnegative, so a discarded loss above the cap can never become feasible. The induction proves both attainability and minimum cost. At the end, no future event uses q, which justifies minimizing over it. Witness updates follow the same legal transitions. □

For two tables over disjoint event domains, define exact min-plus convolution by

$$(P \otimes Q)(\ell, w) = \min_{\substack{\ell_1+\ell_2=\ell \\ w_1+w_2=w}} \{P(\ell_1, w_1) + Q(\ell_2, w_2)\}. \tag{7}$$

Union the witnesses when a minimizing pair is chosen. For a fixed configuration, convolve its child tables, then shift loss by Λ_x and cost by κ_x, and adjoin $D_C(x)$ to the witness. Take the pointwise minimum over configurations *after* each configuration's complete branch table has been formed. Finally convolve the complete branch tables. Loss above R may be discarded throughout.

Correctness part of Theorem 2.1. Every table transition, shift, and convolution constructs a legal deletion set in disjoint original-event domains. Equations (3) and (6) show that its reported counts and cost are exact. Thus every finite output entry has a valid attaining witness.

Conversely, take a minimum-cost feasible deletion set for any count pair. Corollary 3.2 replaces its final-event choices by canonical choices without changing either count and without increasing cost. Each branch's multiplicities occur in its enumerated family by Lemma 3.3. Lemma 4.1 represents every remaining child choice at no higher cost for its exact local counts. Children and branches have disjoint event domains, and every root-ending pair or triple belongs to one branch and one middle-event child. Therefore the corresponding convolutions include the chosen exact global pair. The output cannot cost more than its optimum, while its valid witness prevents it from costing less. Adding F and its fixed cost once completes the result. □

5 Running time, bit complexity, and space

Let $N = M + 1$ and $S = (R + 1)(N_3 + 1)$. Then $S = O(N^5)$ because $R \le M^2$ and $N_3 \le M^3$. A child layer contains at most NS states. Processing an event uses a constant number of transitions per state. A table convolution uses at most S^2 candidate pairs, even without an optimized convolution algorithm. Minimizing over terminal q uses at most NS work per child.

There are at most M events, children, and branches in their respective disjoint collections. Each branch is processed for at most $P_{\le R}$ configurations. Consequently the complete recurrence uses

$$O\Big(P_{\le R}\, N[NS + S^2]\Big) \tag{8}$$

table, arithmetic, and comparison operations, plus polynomial preprocessing and configuration bookkeeping. The enumeration bound in Lemma 3.3 and the sum of the numbers of present groups keep the same single factor $P_{\le R}$. Sorting timestamps and computing suffix counts never enumerate the timestamp

universe. Formula (8) is at most $O(P_{\le R}N^{11})$ operations; it is a conservative bound, not a practical speed estimate.

Every finite cost is a sum of at most M input costs and therefore has $O(L)$ bits. Count indices have $O(\log(M+1))$ bits, and an original-ID witness mask has M bits. On a bit-cost random-access model, the arithmetic, comparisons, and mask operations used here admit an $O(L+1)$ bound per operation. Including polynomial preprocessing gives the deliberately loose bound $O(P_{\le R}(L+1)^{12})$. Other conventional machine models may add a polynomial factor without changing the $2^{O(\sqrt{R})}\,\mathrm{poly}(L)$ conclusion. There is no state dimension or loop up to the monetary budget or total cost. Large costs affect the length of arithmetic, not the number of profile keys.

Process configurations sequentially and discard an individual configuration's working tables after merging it into the branch aggregate. Retain only two child layers, a current child table, current convolution tables, the aggregate branch table, and the aggregate global table. A depth-first configuration generator keeps polynomially many frames. Storing a witness mask in each finite table entry therefore uses polynomial space. Output itself is also polynomially bounded. This completes the complexity proof of Theorem 2.1.

6 Hardness parameterized by the deletion budget

The deletion budget and the preservation slack lead to different parameterized complexity statements. The following companion result holds for the all-optional, unit-cost model. An original event is an individually identified temporal contact; different contacts may have the same endpoints and timestamp. Let E be the event set. For a retained set $S \subseteq E$, the quantity $W_j(S)$ counts j-event walks using events in S, ending at the root, whose timestamps increase strictly. Deleting D therefore leaves count $W_j(E \setminus D)$.

Theorem 6.1 (Hardness in the deletion budget)**.** *Given an explicitly listed temporal in-arborescence of depth at most three, deciding whether*

$$|D| \le b, \qquad W_2(E \setminus D) \ge T_2, \qquad W_3(E \setminus D) \le T_3$$

is W[1]*-hard parameterized by b, even when every original event is optional and has unit deletion cost. The restriction remains hard with a single depth-one vertex and exactly one incoming depth-three event, at timestamp zero, at each depth-two vertex that has such a predecessor.*

The reduction from unweighted bipartite partial vertex cover with side budgets a, q has $b = a + 2q$. Composed with the published reduction from t-Clique, this gives $b = t^2$.

Proof. The source is unweighted bipartite partial vertex cover with two separate side budgets [15, Definition 5 and Theorem 10]. The journal theorem states W[1]-hardness for either side budget; its displayed reduction from regular t-Clique uses $a = t$ and $q = \binom{t}{2}$. Since both budgets are functions of t, that reduction also establishes hardness parameterized by $a + q$. The conference version is [14, Theorem 6]. The regularization in the proof of Corollary A.7 also preserves clique size, so the reduction may start from ordinary Clique. The construction below has three steps. Isolated rows and a pair of columns per source right vertex enforce the two side budgets using the two temporal quotas. Lemma A.4 proves this equivalence when the middle events are temporarily required to survive. Lemma A.5 replaces that requirement by explicitly listed, optional unit-cost events. Lemma A.6 proves polynomial encoded size and the asserted parameter maps. These facts give a parameterized many-one reduction and prove the theorem. □

The complete construction is in Appendix A. Corollary A.7 gives NP-completeness of this decision problem. It is a special case of the profile task: use unit costs, no forced statuses, and select an entry with $w \le T_3$ and cost at most b. The hardness parameter is b; no lower bound in preservation loss R follows.

7 Exact classifier optimization and its boundary

7.1 Predicates of the two counts

Corollary 7.1 (Count-based acceptance)**.** *Let $g(W_2, W_3, z_0)$ be an acceptance predicate, where z_0 is proven invariant under every allowed deletion, including forced deletions. After computing C, minimum-cost acceptance is obtained by selecting the cheapest finite entry with $g(N_2 - \ell, w, z_0) = \text{true}$. If one evaluation takes t_g time, the additional bit time is $O(S(t_g + L + 1))$, including cost comparisons. In particular, a polynomial-time predicate with polynomially encoded model data preserves the parameterized bound, with those encodings included in the input length.*

Proof. All legal deletion sets with a fixed count pair have the same acceptance value. Replacing one by the minimum-cost witness at that pair preserves acceptance. Completeness of the table then reduces the optimum to a scan over its finite entries. A total monetary cap can be checked during the scan. □

For a classifier counterfactual, first compute the factual prediction on the *original history* E, before forced deletions. Set acceptance to the desired class, which must differ from that factual class. A class change already caused by mandatory deletions is legitimate and has the fixed forced cost. The post-forced baseline is an algorithmic normalization, not a new factual observation.

The predicate may be nonmonotone and may be implemented by a decision tree or an ensemble. Its other features must actually be invariant. Replaying one minimum-cost witness per pair through a model with additional dynamic features can establish that witness's feasibility, but not full-model optimality: another witness at the same pair can have different dynamic feature values. A feature computed on the full history also cannot generally be obtained by adding a constant to a windowed walk count, because a deleted event may participate in a walk that crosses the window boundary. The present theorem does not establish exactness for a classifier whose additional inputs change under the allowed deletions.

7.2 An elementary amount-feature obstruction

Binary deletion costs are handled as minimized values. Binary-encoded dynamic features are a different extension, because their attained values can define the acceptance constraint.

Proposition 7.2 (One additive dynamic feature can be hard at $R = 0$)**.** *If acceptance may additionally depend on the sum of retained binary-encoded amounts on optional A events, deciding whether a desired class is attainable is NP-complete even when the collapsed static graph is a single depth-three path, with $R = 0$, unit deletion costs, and a three-leaf classifier whose thresholds are part of the input.*

Proof. Reduce from Subset Sum with positive integers $a_1, \ldots, a_n$ and target k, restricted to $0 < k \le S_a = \sum_i a_i$; trivial targets can be decided before reduction. The standard source problem and its numerical input dependence are discussed by Bringmann [3].

Use four vertices $r \to v \to u \to z$. Create n optional parallel A events $r \to v$ at time zero, one forced-retained B event $v \to u$ at time one, and one forced-retained C event $u \to z$ at time two. Give every optional deletion cost one. Set $T = 1$. Every legal choice has $W_2 = 1$, so $R = 0$. If an upper quota

or budget is desired, $W_3 \le n$ and budget n are nonbinding. Attach amount a_i to optional event i and define the dynamic feature

$$H(D) = \sum_{i:\, A_i \notin D} a_i.$$

The classifier accepts exactly when $H(D) = S_a - k$. A three-leaf tree implements this on integer inputs using thresholds $S_a - k - \frac{1}{2}$ and $S_a - k + \frac{1}{2}$, accepting only the middle interval. Its rational threshold encodings have polynomial length. The original history has $H = S_a$ and is rejected, since $k > 0$.

A deletion set is accepted precisely when its deleted amounts sum to k. The graph, events, amounts, and tree all have polynomial encoding length. Membership in NP follows by checking a deletion witness, its walk counts, the amount sum, and the tree output in polynomial time. □

This is an elementary Subset Sum boundary, not a claim of a new reduction technique. The argument does not establish strong NP-hardness, does not give a hardness result for the precise saved application model, and does not contradict Theorem 2.1. It rules out an unrestricted R-only FPT extension to this extra feature unless P = NP.

8 Worked examples

All examples below use the original root-ending definition. Their standalone JSON files are in the accompanying study artifact's `implementation/examples` directory; the manuscript-source archive contains the typesetting files.

8.1 A nonmonotone target needs the complete profile

On one path, put optional A events at times 0 and 2, both of cost one; force retention of B events at times 1 and 3 and a C event at time 4. Set $T = 2$. The baseline has $(W_2, W_3) = (2, 3)$ and $R = 0$.

W_2	W_3	Minimum cost	Optional deletion witness
2	3	0	none
2	2	1	later A event at time 2
2	1	1	earlier A event at time 0
2	0	2	both A events

Consider a hypothetical classifier whose desired class occurs only at $W_3 = 2$, implemented by thresholds 1.5 and 2.5. The cheapest desired-class change deletes the later A event for cost one. An earliest-A exchange minimizes W_3 for a fixed number of unit-cost deletions, but selects the other event at deletion count one. An algorithm that retains only that minimum-W_3 value loses the accepting state. This example explains the need for the exact w coordinate. It is an illustrative classifier, not a reported result on any trained model.

8.2 Weighted decisions and mandatory deletions

With optional A events at times 0 and 2 of costs 100 and 1, and retained B, C events at times $3, 4$, preserving $W_2 = 1$ gives W_3 values $2, 1, 0$ at minimum costs $0, 1, 101$. The cheapest way to attain $W_3 = 1$ deletes the later event; a unit-cost earliest-event exchange is invalid here.

For the final-event exchange, retain A at time 0 and B events at times $1, 3$, and make two C events at time 4 optional with costs $100, 1$. Both have degree two. With $T = 2$, the attainable pairs are $(4, 4)$ at cost zero and $(2, 2)$ at cost one. Canonicalization must select the cheap final identity, even if an arbitrary identifier order lists the expensive one first.

Finally, retain A at time 0, force deletion of B at time 1 for cost seven, and leave C at time 2 optional with cost three. Set $T = 0$. The original counts are $(1, 1)$, the normalized baseline counts are $(0, 0)$, and the minimum profile has total cost seven with only the mandatory B deletion. The surviving A event retains its original depth-three role. Deleting the now degree-zero C event cannot improve the minimum cost.

9 Related work and parameter transfers

9.1 Temporal reachability, counting, and cut problems

Molter, Renken, and Zschoche [17, Theorem 14] give polynomial-time algorithms for minimizing temporal reachability by deleting or delaying time-edges when the underlying graph is a forest. Their Lemma 2 transfers deletion to delay while preserving the forest property. The stronger weighted forest delay formulation [17, Theorem 15, pp. 10–12] permits nonnegative weights on reached vertices and undelayable time-edges, with a cardinality bound on the modified time-edges. Its dynamic program records a vertex, a delay budget, its first arrival time, and whether that arrival is supplied from inside or outside its rooted subtree; a table entry minimizes the total weight of reached vertices. The underlying temporal graph is undirected, and the forest result allows arbitrary nonnegative traversal times, including the strict convention with traversal time one. Thus tree structure, strictness, and protected contacts alone are not novelty distinctions, and the source's vertex weights must not be confused with arbitrary event-deletion costs.

The final journal version also gives a $4^x|G|^{O(1)}$ algorithm for both reachability problems in the non-strict setting, where x is the timed feedback vertex number [17, Theorem 16]. This parameter is zero on underlying forests. The output contract here instead requires minimum original-event deletion cost at every attainable exact pair of root-ending two- and three-event tuple counts, subject to the preservation quota and shared event decisions. A derivation from the reachability results must therefore supply an encoding or sufficient state for these multiplicities, costs, and constraints, as well as the claimed dependence on preservation loss. Appendix B.2 gives a direct-summary counterexample: the same reached sets and earliest arrivals can coexist with different tuple counts. It does not exclude richer states or auxiliary reductions. The journal article extends the earlier conference version [16]. The comparison above uses the final journal statements and proofs, with their journal theorem numbering.

Enright, Meeks, and Molter [7, Theorem 2] count temporal paths on forests in $O(|V|\tau^2)$ time, where τ is the lifetime. Their Theorems 6–7 treat further structural parameters. Fixed-deletion counting does transfer to the present setting: Appendix B.1 gives a single normalization preserving strict ties, event identities, and both counts under every deletion subset, with lifetime at most M. Thus strictness, orientation, and parallel contacts do not obstruct that counting transfer. The additional task here is to choose one shared deletion set and compute minimum costs for the entire attainable profile with the stated dependence on preservation loss.

Robust Temporal Cut [8] removes contacts so that no source–target temporal path survives even after bounded timestamp perturbations. For undirected temporal graphs, its stated tractability parameters include timed vertex cover, vertex cover plus solution size, and temporal neighborhood diversity plus solution size. This is a related temporal-deletion problem with a different feasibility condition. The existence of those algorithms does not itself supply an exact reduction for two simultaneous multiplicity

counts. Any proposed transfer must also preserve its controlling parameters.

9.2 Cover formulations and the parameters they retain

There is a direct exact set-system representation. Each baseline two-walk is a red element, each baseline three-walk is a blue element, and each optional event supplies the set of elements containing its original identifier. Selecting event sets incurs their deletion costs. A profile entry $C(\ell, w)$ requires red union coverage exactly ℓ and blue union coverage exactly $N_3 - w$. An upper quota on surviving triples instead requires a partial blue target. Shared incidences matter: a walk destroyed by several deletions is still counted once.

This representation relates the problem to red–blue set cover. Abidha and Ashok [1, Theorem 6 in the author preprint] give an exact fixed-dimensional hyperplane algorithm parameterized by red coverage for covering all blue points. The temporal witness incidence admits a three-dimensional coordinate-plane representation, but the compared feasible family has no independent charge for selected planes and freely selects red-free sets. First-event choices in the temporal model can be red-free and still cost money. The partial-blue approximation setting of Chlamtáč, Makarychev, and Vakilian [4, Appendix A] is another close precedent for the benefit–damage formulation; its approximation guarantee does not by itself decide exact profile feasibility and minimum event cost.

Madireddy, Nandy, and Pandit [12, Theorems 1–2] give exact algorithms for intervals and for a restricted axis-parallel-line case. The direct temporal witness representation need not be an interval system. With one first event a, two middle events b_1, b_2, and two final events c_1, c_2, all temporally compatible, the four triples $q_{ij} = (a, b_i, c_j)$ induce the event sets

$$\{q_{11}, q_{12}\}, \quad \{q_{21}, q_{22}\}, \quad \{q_{11}, q_{21}\}, \quad \{q_{12}, q_{22}\}.$$

An interval order would need four distinct adjacent pairs on four blue points, which have only three adjacencies. The restricted line algorithm also relies on disjoint red incidences, whereas a temporal red pair belongs to both its middle and final events. These observations concern the direct representation and do not rule out another exact encoding.

The bipartite partial-cover framework of Mkrtchyan and coauthors [15] includes a selected-vertex cost budget and covered-edge profit targets. Their hardness theorem with two side budgets provides the starting problem for the reduction in Appendix A. For the positive comparisons, neither selected cost nor covered profit is inherently bounded by the preservation loss R. For example, take n optional unit-cost first events before one retained middle/final pair and require preservation of that pair. Then $R = 0$ while destroying all triples requires n deletions. A target leaving at most $\lfloor n/2 \rfloor$ triples makes both the destroyed-triple target and the surviving-triple allowance grow at $R = 0$. An algorithm controlled by those quantities therefore needs an additional parameter argument to yield the bound in Theorem 2.1. This is a statement about the indicated transfer, not about all possible reductions.

9.3 Generalized deletion propagation

Generalized deletion propagation (GDP) permits simultaneous deletion and preservation requirements on multiple query outputs, with shared decisions on source tuples and an objective expressed through view-deletion counts [13, Definition 4.1]. Its ILP framework supports self-joins and both set and bag semantics; its tractability results concern specified deletion-propagation variants and query conditions [13, Sections 5–6].

Appendix E derives a polynomial-size GDP encoding of each unit-cost temporal profile entry. Protected compatibility relations compile strict timing and root incidence; retaining every event identifier in the

query output preserves tuple multiplicity under set semantics. Deletion and preservation bounds on the same query enforce an exact count. Mandatory statuses are handled by normalization and singleton preservation queries. Polynomially many target pairs recover the full profile table. This is a derived encoding, not a temporal theorem stated in the cited work.

The appendix also gives a weighted witness-indicator ILP with the original binary cost coefficients. That objective extends the literal cardinality objective of GDP Definition 4.1; expanding an event into one copy per cost unit would not justify polynomial input size. Thus the formulation and full-table output are not the claimed novelty. The contribution claimed in Theorem 2.1 is the structural algorithm with its joint time and space guarantee. Obtaining that guarantee through GDP requires a solving-time and parameter-transfer argument beyond the encoding.

9.4 General representations and optimization frameworks

Courcelle and Durand's fly-automata framework computes satisfying assignments, cardinality spectra and related functions (Theorem 23), and develops minimizing algebras and weighted cardinality optimization in Section 3.2.3 [5]. It is a direct precedent for retaining sufficient attainable information and optimizing within states. Its complexity discussion requires bounds on the chosen automata and representations. A transfer needs an explicit representation preserving shared event decisions and exact counters, together with bounds implying the stated running time. The quotient and enumeration in Sections 3–4 provide one such construction. A suitable application to this class could provide another presentation of the algorithm.

Del Pia proves polynomial solvability for factorized binary polynomial optimization with fixed structural parameters, with explicit dependence on the factorization in Theorems 1–2 and a conversion result in Proposition 2 [6]. This framework can express the weighted profile search. The issue is the parameter transfer, not expressibility. The following direct encoding makes that distinction concrete.

For an exact target pair (p, q) and a trial optional monetary cap b, use one Boolean deletion variable per optional event. Substitute forced statuses and express $W_2(x)$ and $W_3(x)$ as sums of retained-pair and retained-triple monomials in $(1 - x_e)$. Let $C_O = \sum_{e \in O} c_e$, take $0 \le b \le C_O$, and encode a nonnegative slack σ using enough binary digits to represent every integer through C_O (no digits are needed if $C_O = 0$). Maximize

$$-\left[(W_2(x) - p)^2 + (W_3(x) - q)^2 + \left(b - \sum_{e \in O} c_e x_e - \sigma\right)^2\right]. \tag{9}$$

Its maximum is zero exactly when the pair is attainable within cap b. After Boolean multilinearization, the degree is at most six and the expanded representation has polynomial size and coefficient bit length. First test $b = C_O$ to detect an unattainable coordinate. For an attainable one, binary search over b requires only $O(1+\log(C_O+1))$ calls, including the zero-cost case. If a polynomial format omits constants, carry the known offset into the comparison threshold; constant or purely linear cases can be solved directly. The encoding above is a derivation for the temporal problem; the cited paper does not state this transfer. A fixed degree does not by itself give a bounded number of factors. The cited runtime's factor-dependent input exponents therefore do not, without another bound, yield $2^{O(\sqrt{R})}\,\mathrm{poly}(L)$ for this transfer.

Kundu and coauthors' Theorem 1.7 concerns constrained, class-based MaxSAT with a true-variable budget and a randomized approximation guarantee [11]. Unit-cost quota encodings can introduce event-deletion variables and pair-loss variables; their budget includes both the deletion allowance and R. For integral class totals, sufficiently small approximation error makes a successful returned solution meet the original targets exactly, retaining the theorem's success probability of at least $1 - 1/e$. The resulting dependencies need not be bounded by R: the deletion allowance and objective target can grow already

at $R = 0$. The weighted exact-profile extension also needs an explicit monetary constraint encoding. The unit-cost budget argument cannot simply be relabeled as arbitrary event-cost support.

Khouna and coauthors formulate optimal tree-ensemble counterfactual search using a desired prediction, a cost, and plausibility/actionability constraints (Section 2, Equation 1), then compare exact solution paradigms [10]. Thus targeting an actual class and minimizing a change cost are not new objectives. The classifier corollary applies the specialized temporal-profile algorithm only when the model depends on the supported counts and invariant inputs. It does not introduce desired-class optimization or a general framework for counterfactual search.

9.5 Scope of the comparison

The inspected results supply important counting, representation, and optimization tools. The GDP construction in Appendix E establishes a semantic transfer for the unit-cost formulation and gives an explicit weighted ILP extension. It does not analyze the time needed to solve those encoded instances as a function of preservation loss. An algorithmic transfer yielding Theorem 2.1 must preserve the shared original-event decisions and exact output contract, and its representation size and parameter dependence must together imply $2^{O(\sqrt{R})}\,\mathrm{poly}(L)$ time and polynomial space, including the complete input encoding and an absolute polynomial exponent.

The direct proof supplies this guarantee for the stated temporal class. The comparison does not prove that no alternative derivation or earlier result can imply the theorem. Exact profiles, weighted minima, integer partitions, min-plus convolution, and terminal predicate filtering are not claimed as new general techniques.

10 Implementation and validation

10.1 Comparison with exhaustive enumeration

For each small test instance, an exhaustive checker enumerates ordered pairs and triples of original events, requiring matching endpoints, strictly increasing timestamps, and a final endpoint at the root. For every legal deletion set, it counts the surviving tuples without calling the solver's counting, grouping, or optimization routines. At each quota it compares the attainable count pairs and their minimum costs with those returned by the solver, and checks every deletion witness directly. Tied optimal witnesses need not use the same event identifiers.

The frozen corpus uses seed 20261104. It contains 20 named histories, all $3^5 = 243$ timestamp assignments from $\{0, 1, 2\}$ on a fixed five-event path, 80 seeded random histories, and 32 transformed histories. The transformations are input permutation, injective identifier renaming, order-preserving affine timestamp transformation, and positive cost scaling on eight selected histories. Named cases include zero costs, strict ties, parallel events, forced disconnection, multiple branches, and unequal costs within equal-degree final-event classes.

Check	Completed
Frozen histories	375
Quota queries / valid-input solver calls	1,863
Feasible / infeasible quota queries	1,062 / 801
Queries with $R = 0$	561
Legal deletion subsets enumerated	81,108
Exact profile entries and witnesses checked across queries	2,529
Acceptance/budget-selection queries	54,076
Returned selection witnesses checked	19,731
Invalid inputs correctly rejected	57
Transformation relations checked	32

Selection tests use nine predicates, including intermediate exact counts, parity and disconnected count regions, and monetary caps at the optimum and adjacent values. Witness verification checks original identities, forced statuses, optional and total costs, both root-ending counts, and the quota. The largest history has 12 events and costs of up to 203 bits. Maximum baseline counts are $W_2 = 21$ and $W_3 = 6$. The profile total counts checks across quota queries, not necessarily distinct global coordinates.

10.2 Compatibility and native-run evidence

The original full suite passed on CPython 3.12.14 on Linux. A subsequent compatibility repair removed Python-version-sensitive dataclass and evaluated type-alias syntax. All 14 solver function abstract syntax trees were unchanged. Smoke checks covered CPython 3.8.20 and 3.9.25 on Linux, and the full frozen suite passed on CPython 3.9.25 with identical reference output digests.

The macOS validation summary records a successful run of the same suite on 4 November 2026 using CPython 3.9.6 on macOS 27.0.1 (ARM64). The SHA-256 hashes of the solver, validator, corpus, and protocol match those of the compatibility-corrected release. The output hashes recorded in the summary match those computed from the retained reference files. For this suite, the archive contains the macOS summary but not the full macOS outputs; those files could not be hashed separately.

The implementation requires Python 3.8 or newer and the standard library. These are finite correctness and compatibility checks. No runtime-performance comparison is inferred from their elapsed times. The general result rests on the proof. The implementation archive also retains the initial root-endpoint specification correction, both frozen oracle versions, and their audit records; Appendix D explains the provenance.

The ten-event illustration in Proposition 3.4 is archived separately with its stated expected profiles and a direct enumeration of all 64 legal optional-deletion subsets. It is an explanatory check added with this manuscript and is not counted in the original 375-history suite.

10.3 Exact targets in synthetic transaction histories

The file `HI-Small_Trans.csv` from version 8 of IBM's synthetic anti-money-laundering dataset [2] contains 5,078,345 transactions. The sampling unit was a destination account and transaction minute. An earlier audit fixed 1,024 queries by seeded hash order from the 3,854,758 distinct units with a full 24-hour lookback, without using labels or structural eligibility. These queries contain 1,023 distinct roots. Each context includes every transaction in $[t - 1440, t)$ between accounts within incoming static distance three of the root in that window. The extraction retained all such events, including loops, tied times and parallel contacts; no conflicting edge was removed to make a context eligible. The cohort had

been studied previously, and queries may share events. The results describe these queries rather than independent trials or an unseen test set.

Coverage. All 1,024 contexts were extracted and their baseline counts determined. The structural check was applied to the original induced context. Table 1 retains empty, unsupported, and resource-limited queries in the denominator. Of 232 queries with baseline $W_3 > 0$, 77 fit the model and 68 received complete profiles. The nine skipped eligible positive queries had a median of 91 events compared with 12 for the 68 profiled queries.

Query disposition	$W_3 = 0$	$W_3 > 0$	Total
Empty context	179	0	179
Outside the required tree structure	125	155	280
Eligible, above the 32-event cap	34	9	43
Eligible, all requested profiles completed	454	68	522
Total	792	232	1,024

Table 1. Disposition of all sampled account/minute queries.

Two questions asked of the same profile. Every event was optional with unit deletion cost. Given baseline counts N_2, N_3, set $T = \lceil pN_2/100 \rceil$ for $p \in \{100, 95, 90\}$ and $U = \lfloor 3N_3/4 \rfloor$. For each T, the evaluation compared the goals $W_3 \leq U$ and $W_3 = U$, both subject to $W_2 \geq T$. Budgets 0, 1 and 3 were applied to the minimum deletion cost. The 454 profiled queries with $N_3 = 0$ satisfy both goals without intervention; the comparison below therefore concerns the 68 profiled queries with $N_3 > 0$.

	Attainable within three deletions		Unattainable at any cost
Preservation	$W_3 \leq U$	$W_3 = U$	$W_3 = U$
100%	68	13	55
95%	68	14	54
90%	68	15	53

Table 2. Each row refers to the same 68 positive, fully profiled queries.

At full preservation, the least cost for the upper bound was one deletion in 59 queries, two in eight, and three in one. In 55 of the 68, no deletion set could attain the exact same target while preserving every pair, even without a deletion budget. At $R = 0$, every middle or final event belonging to a baseline pair must survive. Deleting a first event then removes all triples extending it as one group; different first events remove disjoint groups. Attainable triple losses are sums of these group sizes, so some intermediate integer losses can be absent. For example, in the preidentified pilot query `q0011`, preserving seven pairs permits either two triples at cost zero or zero triples at cost one. Exactly one triple is unattainable. Rounding also matters: on 50 of the 68 positive queries all three percentages require the same integer T. The rows of Table 2 are paired comparisons, not 204 independent observations.

Verification and scope. Deduplicating equal values of T gave 653 planned tables among the 565 eligible nonempty queries. The 32-event cap excluded 90 tables before optimization; these have unknown outcomes. There were no solver or oracle timeouts. The Mac run reported witness replay for all 935 returned entries across the 563 completed tables. For 528 of those tables (at most 16 optional events), exhaustive enumeration agreed on every attainable pair and its minimum cost. The other 35 had witness

replay but no exhaustive comparison. The run provides evidence about local count sensitivity and model coverage on synthetic transactions. It does not measure classification accuracy, changes in fraud decisions, or comparative runtime.

11 Discussion and conclusion

Preservation loss bounds the structural choices of deleted final events even when the deletion budget, cost magnitudes, and number of branches are not parameter-bounded. The continuation equivalence of final events makes those choices enumerable by integer partitions. Conditional chronological states then recover exact minimum-cost profiles, and branchwise aggregation avoids multiplying configuration families. The full proof yields an absolute polynomial exponent and polynomial space, including original-event witnesses.

The GDP encoding clarifies the scope of this result. Simultaneous exact count requirements and a complete table of count optima fit a broader optimization framework; the structural state, charging argument, and preservation-loss complexity bound are the contribution proved here. The weighted ILP formulation provides another exact representation of the same feasible event choices. Neither that encoding nor the fractional example in Appendix E rules out stronger formulations or another derivation through existing methods.

The budget-hardness result concerns a different parameter and establishes no tight lower bound for the square-root dependence on R. The algorithm's polynomial factor is deliberately conservative; the finite checker is a correctness artifact rather than an efficiency experiment. A practical comparison with a generic exact ILP baseline should use the same event identities, costs, forced statuses, and full profile output, with controlled variation in R. Such a runtime comparison is not part of the present evaluation.

The supported model has explicit events, fixed depth at most three, strict timestamps, nonnegative deletion costs, and root-ending walk counts. Negative costs, compressed event multiplicities, deeper temporal structures, or extra dynamic feature constraints require separate analysis. An independent cardinality cap in addition to arbitrary costs is not encoded in the current table. These boundaries leave precise questions for further work while keeping the present theorem and its classifier corollary fully specified.

A Complete construction for budget hardness

Source problem. An instance of two-budget bipartite partial vertex cover consists of a simple bipartite graph $H = (X, Y, E_H)$, nonnegative integer budgets a, q, and an integer target ℓ. It asks for sets $X' \subseteq X$ and $Y' \subseteq Y$, with $|X'| \leq a$ and $|Y'| \leq q$, such that at least ℓ edges have an endpoint in $X' \cup Y'$. Write $n_0 = |X|$, $m = |Y|$, and $e = |E_H|$. The source is unweighted; the two budgets are separate cardinality caps.

The regular-Clique source map. The following paragraphs detail the preceding reduction in the cited article and its boundary cases. Let G be a simple d-regular graph with N vertices, m_G edges, and nonnegative Clique target t. Cases $t < 3$, $t > N$, and $t = N$ are decidable in polynomial time; the last case tests completeness. Cases $d < t - 1$ or $m_G < \binom{t}{2}$ are no-instances. Map every directly decided case to the three-event temporal yes/no template below, carrying the numerical budget $b = t^2$. It remains to consider $3 \leq t < N$, $d \geq t - 1$, and $m_G \geq \binom{t}{2}$.

Make one vertex of X for each vertex of G, and one vertex of Y for each edge of G. Join $v \in X$ to $g \in Y$ exactly when v is not an endpoint of g. This non-incidence graph has $N + m_G$ vertices and

$e = (N-2)m_G$ edges. Set

$$a = t, \qquad q = \binom{t}{2}, \qquad \ell = t(m_G - d) + (N-t)q.$$

Any source selection within these caps can be extended to exactly t left vertices and q right vertices without losing covered edges. For these saturated choices $S \subseteq V(G)$ and $J \subseteq E(G)$, let $i(S,J) = \sum_{g\in J} |\,\mathrm{endpoints}(g) \cap S|$. Inclusion–exclusion gives the exact covered-edge count

$$\begin{aligned} C(S,J) &= t(m_G - d) + q(N-2) - \big(tq - i(S,J)\big) \\ &= \ell - \big(2q - i(S,J)\big). \end{aligned} \tag{10}$$

Since $i(S,J) \le 2q$, reaching ℓ is equivalent to every edge of J having both endpoints in S. A simple graph on t vertices has at most $q = \binom{t}{2}$ edges, so every such S is a clique. Conversely, a t-clique and its internal edges attain equality. This is the reduction of [15, Theorem 10], now with its at-most-to-exact step and coverage identity explicit. The construction is polynomial, both side budgets depend only on t, and the temporal map below gives $b = a + 2q = t^2$. If the resulting partial-cover target exceeds e, the following preprocessing correctly returns a no-instance.

Trivial partial-cover instances. The cases $\ell \le 0$ and $\ell > e$ are immediate. If $a \ge n_0$ or $q \ge m$, selecting an entire side covers every edge. If $q = 0$, choose the a largest left degrees, or all left vertices when fewer are available: edges incident to different left vertices are distinct, so this maximizes coverage. Thus the construction may assume

$$0 \le a \le n_0, \qquad 1 \le q < m, \qquad 0 \le \ell \le e. \tag{11}$$

To preserve the exact formula $b = a + 2q$ on the directly decided cases as well, use the three-event path $s \to v \to u \to z$, at times $0, 1, 2$, with the same numerical budget $b = a + 2q$. Set $(T_2, T_3) = (1, 1)$ for a yes-instance and $(T_2, T_3) = (2, 1)$ for a no-instance. The first accepts $D = \varnothing$; the second is impossible because its undeleted two-walk count is one. Every event in either instance is optional.

Rows, numerical constants, and columns. Add $h = 2m+1$ isolated left vertices to H, and number the resulting rows $1, \dots, n$. Define

$$\begin{aligned} &n = n_0 + 2m + 1, && r = n - a, && b = a + 2q, \\ &\kappa = e + 1, && P = 4m^2 + 1, \\ &\Lambda = \big((\kappa+1) + r\kappa\big)mP + e + 1. \end{aligned} \tag{12}$$

In particular,

$$\begin{aligned} r > 2m, \qquad \kappa > e, \qquad P > 2m^2, \\ \Lambda + 1 > (\kappa+1)mP, \qquad \Lambda > r\kappa mP + e. \end{aligned} \tag{13}$$

The isolated rows have no source edges and require no additional budget.

For each $y_j \in Y$, introduce an original column of numerical weight j and a companion column of numerical weight $P - j$. All $2m$ weights are distinct: the original weights are $1, \dots, m$, and $P - m > m$. Sort the columns by increasing weight and write their weights as $\rho_1 < \cdots < \rho_{2m}$. Their sum is mP. These weights specify event counts in the construction; they are not event deletion costs.

For row i and column t, let N_{it} be the adjacency indicator of the padded graph when column t is original. Set $N_{it} = 0$ for companion columns and for padded rows. Let $\sigma_t = \sum_{i=1}^{n} N_{it}$; hence $0 \le \sigma_t \le n_0$.

The intermediate temporal instance. Use the static arcs

$$s_i \longrightarrow v_i \longrightarrow u \longrightarrow z \quad (1 \le i \le n), \qquad v_0 \longrightarrow u.$$

The root is z. There is a single depth-one vertex u; the auxiliary depth-two vertex v_0 has no predecessor. For each row i, put one event $A_i : s_i \to v_i$ at time zero. For every sorted column t, put one event $C_t : u \to z$ at time $2t$.

Prescribe the following cumulative numbers of middle events strictly before C_t:

$$\mu_{it} = \Lambda + \kappa\rho_t + N_{it} \quad (1 \le i \le n), \qquad \mu_{0t} = n + n\rho_t - \sigma_t. \tag{14}$$

Set $\mu_{i0} = 0$ for $0 \le i \le n$. For each t, put exactly $\mu_{it} - \mu_{i,t-1}$ distinct events on $v_i \to u$, all at time $2t-1$, for every $0 \le i \le n$. Call these the B-events.

All these multiplicities are nonnegative integers. The initial prefixes are positive. For consecutive columns after the first, their increments are bounded below by

$$\kappa(\rho_t - \rho_{t-1}) + (N_{it} - N_{i,t-1}) \ge \kappa - 1 \ge 0$$

on a productive row, and by

$$n(\rho_t - \rho_{t-1}) - (\sigma_t - \sigma_{t-1}) \ge n - n_0 > 0$$

on the auxiliary row. The times $2t-1$ and $2t$ realize precisely the prefixes in (14), with strict temporal inequalities. All contacts, including contacts sharing endpoints and a timestamp, receive distinct IDs.

For the moment, B-events must survive, while A- and C-events are optional. This is an intermediate construction only; the final instance will make every event optional.

Lemma A.1 (Exact counts in the intermediate instance)**.** *Let $I \subseteq \{1,\dots,n\}$ be the retained A-rows and $J \subseteq \{1,\dots,2m\}$ the retained C-columns. Put*

$$p = |I|, \qquad s = |J|, \qquad \omega = \sum_{t\in J} \rho_t, \qquad N(I,J) = \sum_{\substack{i\in I\\ t\in J}} N_{it}.$$

When all B-events survive, the counts and deletion cost are

$$\begin{aligned} W_2^\circ(I,J) &= n\big[s(\Lambda+1) + (\kappa+1)\omega\big], \\ W_3^\circ(I,J) &= p\big[s\Lambda + \kappa\omega\big] + N(I,J), \\ |D_{AC}| &= (n-p) + (2m-s). \end{aligned} \tag{15}$$

Proof. Each two-walk ending at z consists of a B-event followed by a retained C_t. Its contribution at column t is

$$\sum_{i=1}^{n} \mu_{it} + \mu_{0t} = n\Lambda + n\kappa\rho_t + \sigma_t + n + n\rho_t - \sigma_t = n\big[(\Lambda+1) + (\kappa+1)\rho_t\big].$$

An A-event cannot belong to a two-walk ending at the root. Each three-walk consists of A_i, B, C_t, with $i \in I$, $t \in J$, and the B-event counted by μ_{it}. The unique A_i precedes every B-event on its row. The auxiliary row contributes no three-walks. Summing μ_{it} over $I \times J$ proves the second identity. The last identity counts the deleted optional IDs. □

Define the target retained-column count and target weight by

$$s_\star = 2(m-q), \qquad \omega_\star = (m-q)P.$$

They satisfy $s_\star \geq 2$, $\omega_\star > 0$, and $n + 2m - b = r + s_\star$. Set the intermediate quotas to

$$\begin{aligned} T^\circ &= n\big[s_\star(\Lambda+1) + (\kappa+1)\omega_\star\big], \\ U^\circ &= r\big[s_\star\Lambda + \kappa\omega_\star\big] + e - \ell. \end{aligned} \tag{16}$$

Lemma A.2 (The two quotas enforce the budget allocation)**.** *An A/C deletion set of cost at most b satisfies $W_2^\circ \geq T^\circ$ and $W_3^\circ \leq U^\circ$ if and only if*

$$p = r, \qquad s = s_\star, \qquad \omega = \omega_\star, \qquad N(I,J) \leq e - \ell. \tag{17}$$

Every feasible intermediate deletion set has cost exactly b.

Proof. Suppose the two quotas hold. Since $0 \leq \omega \leq mP$, if $s < s_\star$ then

$$\frac{W_2^\circ - T^\circ}{n} = (s - s_\star)(\Lambda+1) + (\kappa+1)(\omega - \omega_\star) \leq -(\Lambda+1) + (\kappa+1)mP < 0,$$

contradicting the lower quota. Therefore $s \geq s_\star$.

Suppose instead that $s = s_\star + d$, where $d \geq 1$. The deletion budget gives $p + s \geq r + s_\star$, so $p \geq r - d$. Since $d \leq s \leq 2m < r$, the latter lower bound is positive. Moreover,

$$ps \geq (r-d)(s_\star + d) = rs_\star + d(r-s) \geq rs_\star + 1.$$

All terms in the formula for W_3° are nonnegative. It follows that

$$\begin{aligned} W_3^\circ &\geq (rs_\star + 1)\Lambda \\ &> rs_\star\Lambda + r\kappa mP + e \\ &\geq rs_\star\Lambda + r\kappa\omega_\star + e - \ell = U^\circ, \end{aligned}$$

a contradiction. Consequently $s = s_\star$.

The budget now gives $p \geq r$, and the lower quota gives $\omega \geq \omega_\star$. Subtracting the base term of U° from the triple count yields

$$\begin{aligned} W_3^\circ - r(s_\star\Lambda + \kappa\omega_\star) = (p-r)(s_\star\Lambda + \kappa\omega_\star) \\ + p\kappa(\omega - \omega_\star) + N(I,J). \end{aligned}$$

All three terms on the right are nonnegative, and their sum is at most $e - \ell \leq e$. If $p > r$, the first term is at least $s_\star\Lambda \geq 2\Lambda > e$. If $\omega > \omega_\star$, integrality makes the second term at least $r\kappa > e$. Thus $p = r$, $\omega = \omega_\star$, and the remaining inequality is $N(I,J) \leq e - \ell$. Conversely, substituting (17) into (15) verifies both quotas. The same substitution gives cost $(n-r) + (2m - s_\star) = a + 2q = b$. □

Lemma A.3 (Companion balance)**.** *If $s = s_\star$ and $\omega = \omega_\star$, exactly $m - q$ original columns and $m - q$ companion columns survive.*

Proof. Let c be the number of retained companion columns. Their weights and the original-column weights give

$$\omega = cP + \delta, \qquad \delta = \sum_{\substack{j:\ \text{original} \\ \text{survives}}} j - \sum_{\substack{j:\ \text{companion} \\ \text{survives}}} j.$$

There are at most $2m$ summands, each of absolute value at most m, so $|\delta| \leq 2m^2 < P$. As $\omega = (m-q)P$, the integer $(m - q - c)P = \delta$ can only be zero. Hence $c = m - q$; the remaining $s_\star - c = m - q$ retained columns are original. The argument does not require matching identities for retained originals and companions. □

Lemma A.4 (Equivalence with retained middle events)**.** *The source partial-cover instance is feasible if and only if the intermediate temporal instance admits a deletion set of cost at most b, with $W_2^\circ \geq T^\circ$ and $W_3^\circ \leq U^\circ$.*

Proof. First take a source cover X', Y'. Extend its selected left vertices to exactly a rows of the padded graph, and extend Y' to exactly q right vertices. Both extensions are possible and preserve coverage. Delete the corresponding a events A_i. For every selected right vertex y_j, delete its original C-event and its matching companion C-event. This uses exactly $a + 2q$ deletions. The retained columns consist of $m - q$ pairs, each of weight P. Thus $p = r$, $s = s_\star$, and $\omega = \omega_\star$. The quantity $N(I, J)$ is exactly the number of source edges not covered by the extended selections, so it is at most $e - \ell$. Lemma A.2 gives the two temporal quotas.

Conversely, consider a feasible intermediate deletion set. Lemma A.2 gives exactly a deleted A-events, and Lemma A.3 gives exactly q deleted original C-events and q deleted companion C-events. Select the original left vertices whose A-events were deleted, omitting any deleted padded rows, and select the right vertices whose original C-events were deleted. These selections use at most a left vertices and exactly q right vertices. Only original columns and original rows can have nonzero N_{it}. Therefore $N(I, J)$ counts precisely the source edges with neither endpoint selected. Since $N(I, J) \leq e - \ell$, the selections cover at least ℓ edges. □

Removing every retention requirement. Set

$$Q = 2mb + 1, \qquad T_2 = QT^\circ, \qquad T_3 = QU^\circ. \tag{18}$$

Replace each intermediate B-event by Q distinct optional events with the same endpoints and timestamp. Keep the A- and C-events unchanged, and retain budget b. All final events are original input IDs with unit deletion cost. The intermediate instance is used only to prove this final construction; it imposes no retention restriction on the final input.

Lemma A.5 (Optional replication)**.** *The all-optional final instance is feasible if and only if the intermediate instance is feasible. In fact, every feasible final deletion set deletes no B-event.*

Proof. Fix the A/C choices of an arbitrary final deletion set D with $|D| \leq b$, and let w_j be the intermediate count with those choices and all middle events retained. Restoring every deleted B copy in the final instance gives exactly Qw_j surviving j-walks, for $j = 2, 3$.

A B copy belongs to at most $2m$ surviving three-walks: its row has at most one A-event, and there are at most $2m$ final C-events. An auxiliary copy belongs to none. Each triple contains exactly one B copy, so the triple loss caused by all deleted B copies is at most

$$2m\,|D \cap B| \leq 2mb = Q - 1.$$

If $w_3 > U^\circ$, integrality would therefore imply

$$W_3(E \setminus D) \geq Q(U^\circ + 1) - (Q - 1) = QU^\circ + 1 > T_3,$$

contrary to feasibility. Hence $w_3 \leq U^\circ$. Restoring middle events can only increase the two-walk count; consequently

$$Qw_2 \geq W_2(E \setminus D) \geq QT^\circ,$$

and $w_2 \geq T^\circ$. The A/C part of D alone costs at most b, so it is a feasible intermediate solution. By Lemma A.2, that part already uses exactly b deletions. Thus D contains no deleted B copy.

Conversely, a feasible intermediate A/C deletion set can be used in the final instance while deleting no B copy. The two counts are then exactly Qw_2 and Qw_3, and both final quotas hold. □

Lemma A.6 (Encoded size and parameter map)**.** *The nontrivial construction contains exactly*

$$M = n + 2m + Qn[\Lambda + (\kappa+1)P - \kappa] \tag{19}$$

explicitly listed events, uses timestamps in $\{0,\dots,4m\}$*, and has size polynomial in the explicit source input length. Its deletion budget is* $b = a + 2q$*, and the composition with the* t*-Clique reduction in [15, Theorem 10] has* $b = t^2$*.*

Proof. The last column has weight $P-1$ and is a companion; hence all its adjacency indicators and its σ-value are zero. Each productive row contains $\Lambda + \kappa(P-1)$ intermediate B-events, and the auxiliary row contains nP. Their total is

$$n[\Lambda + \kappa(P-1)] + nP = n[\Lambda + (\kappa+1)P - \kappa].$$

Multiplying by Q and adding n events of type A and $2m$ of type C gives (19). The static tree has $2n+3$ vertices and depth three. Its only depth-one vertex is u. Replication changes neither the static tree nor any timestamp.

Put $S_H = n_0 + m$. In the range (11),

$$e = O(S_H^2), \quad n, r, b = O(S_H), \quad \kappa, P, Q = O(S_H^2),$$
$$\Lambda = O(S_H^6), \quad M = O(S_H^9).$$

Moreover $T^\circ, U^\circ = O(S_H^8)$, so $T_2, T_3 = O(S_H^{10})$. Event IDs, endpoints, timestamps, budgets, and quotas therefore need only $O(\log S_H)$ bits per numerical field. In particular, an explicit event list has length $O(S_H^9 \log S_H)$. Every contact is written out; the construction uses no binary-encoded multiplicities. The directly decided cases have constant-size temporal event lists and retain the binary encoding of $b = a + 2q$.

The parameter bound $b = a + 2q \le 2(a+q)$ is independent of n_0, m, e. The source theorem uses budgets $a = t$ and $q = \binom{t}{2}$ in its reduction from regular t-Clique. Thus the composed budget is

$$b = t + 2\binom{t}{2} = t^2.$$

This identity also holds if a directly decided source case uses the three-event template described above. □

Why the preservation-slack algorithm is consistent with this result. For the nontrivial construction, the final instance has no mandatory deletions and $T_2 > 0$. Its preservation slack $R = W_2(E) - T_2$ is exactly

$$R = Qnq[2(\Lambda+1) + (\kappa+1)P]. \tag{20}$$

Indeed, the full retained column set has $s = 2m$ and weight mP; subtracting QT° from the first identity in (15), scaled by Q, gives (20). This quantity is unbounded in the source graph size even with fixed budgets. For example, fix $a = q = 1$, $m = 2$, $e = \ell = 1$, and increase the number of isolated left vertices. Then $b = 3$ is fixed, whereas n, Λ, and R grow. Thus this reduction does not bound R by a function of b. It is consistent with a $2^{O(\sqrt{R})}\,\mathrm{poly}(|\mathcal{I}|)$ algorithm parameterized by preservation slack. The conclusion is W[1]-hardness in b; no W[1]-membership claim or ETH exponent lower bound is asserted.

A.1 Classical complexity consequence

The explicit construction also yields the following classical consequence. The proof is included for completeness, without a historical-priority claim for this corollary or the elementary regularization used below.

Corollary A.7 (Classical NP-completeness)**.** *The decision problem in Theorem 6.1 is NP-complete, even with all events optional and of unit cost, one depth-one vertex, and exactly one time-zero A event at each productive depth-two vertex.*

Proof. Membership in NP follows from an original-event deletion bit vector. Its budget and forced statuses, when present, can be checked directly. Enumerating ordered retained event pairs and triples, and testing their endpoints and strict timestamp inequalities, computes the required counts using $O(M^3)$ candidate tuples. Endpoint and timestamp comparisons, comparison with the encoded quotas, and validation of the structural restrictions all take polynomial time in the complete input length.

Classical hardness follows by a polynomial chain of reductions from ordinary Clique. Let G have N vertices and maximum degree Δ, with target t. Cases $t < 3$, $t > N$, or $\Delta = 0$ are directly decidable. Otherwise make 2Δ disjoint copies of G, named L_i, R_i for $i \in \mathbb{Z}_\Delta$. For every original vertex v of degree $d(v)$, join its copy in L_i to its copies in R_{i+j}, for $0 \le j < \Delta - d(v)$, with indices modulo Δ. Each copied vertex receives exactly its degree deficit, on both sides, so the resulting simple graph is Δ-regular. No added edge lies in a triangle: its endpoints have the same original label in different copies; a vertex of another label can be an internal neighbor of at most one endpoint, while all edges among copies of the same label are bipartite. Thus every clique of size at least three remains within an original copy, preserving the existence of a t-clique. The regularization has $2N\Delta = O(N^2)$ vertices and $N\Delta^2 = O(N^3)$ edges and is polynomial.

The regularized graph has $N' = 2N\Delta > t$, so the strict source-size condition is automatic. Apply the regular-Clique prechecks and source-map argument above. For its N' vertices, m' edges, and degree d, the non-incidence construction in [15, Theorem 10] uses one source-side vertex per vertex and one per edge, with

$$a = t, \qquad q = \binom{t}{2}, \qquad \ell = (N' - t)\binom{t}{2} + t(m' - d).$$

The graph has $N' + m'$ vertices and at most $N'm'$ edges, so this step is polynomial. Apply the temporal construction above. Its side-size sum satisfies $S_H = O(N^3)$; hence Lemma A.6 gives polynomial encoded output size, even with the loose event bound $O(N^{27})$. Directly decided source cases use the stated constant-event yes/no templates. The output has the promised structural restrictions, and its budget is $b = t + 2\binom{t}{2} = t^2$. This polynomial many-one reduction proves NP-hardness and completes the proof. □

Consequently, an exact polynomial-time algorithm for every instance of this full model would imply P = NP. This consequence does not rule out a simpler or faster algorithm parameterized by preservation slack R, and it supplies no matching lower bound for the square-root exponent.

B Counting-model transfer and a reachability-summary distinction

B.1 A simultaneous normalization for every deletion subset

The forest counting theorem of Enright, Meeks, and Molter [7] uses undirected temporal paths with nondecreasing timestamps. The following explicit transformation explains why fixed-deletion counting

transfers to the present directed, strict, parallel-event convention. The transformation below establishes the comparison; it is not asserted as a theorem stated in the cited work.

Proposition B.1 (Deletion-preserving counting normalization)**.** *For an explicit depth-three temporal in-arborescence with $M > 0$ events, there is a polynomial-time transformation to a temporal forest with at most M distinct time labels such that, simultaneously for every retained event subset, the number of temporal paths from vertices of original depth h to the root is exactly W_h, for $h = 2, 3$. Original event identities are preserved bijectively.*

Proof. Let $\rho(e)$ be the rank of $t(e)$ among distinct original timestamps, preserving ties. Let $d(e) \in \{1, 2, 3\}$ be the original depth of the source of e, and give each original event a distinct index $i(e) \in \{0, \dots, M-1\}$. Define

$$t'(e) = 4M\rho(e) + Md(e) + i(e).$$

For consecutive rootward events e, f, their depths differ by one, so

$$t'(e) - t'(f) = 4M(\rho(e) - \rho(f)) + M + i(e) - i(f).$$

The final term lies between 1 and $2M - 1$. Hence $t(e) < t(f)$ if and only if $t'(e) \le t'(f)$. Equal original timestamps are incompatible in the rootward direction. All transformed labels are distinct: different ranks are separated by the $4M$ term, different depths within a rank by M, and different identities within a rank and depth by their indices. Rank-compressing these labels preserves their order and leaves at most M time labels.

Forget arc directions. The underlying graph is a tree, so the unique simple path from any vertex to the root follows the original rootward direction. A temporal path cannot leave that route and return without repeating a vertex. Its successive transformed labels are nondecreasing exactly when the original labels are strictly increasing. Distinct original parallel contacts remain distinct time-edges. The correspondence survives deletion of any common subset of identities and preserves the number of events in each path. Summing over starting vertices of original depth h gives the claimed equality. □

Empty histories are handled directly. Forced statuses and costs may be carried as identity metadata, but the cited counting theorem does not optimize them. Counting for each fixed deletion set therefore transfers; enumerating and optimizing a shared deletion set still requires a separate argument. This normalization concerns simple temporal paths on a forest, not arbitrary undirected walks that may return along an edge.

B.2 Reached sets and earliest arrivals do not encode tuple quotas

Consider the path $a \to v \to u \to z$, with three optional first events at times $1, 4, 7$, three retained middle events at times $2, 5, 8$, and a retained final event at time 10. There are three two-walks and six three-walks. Deleting the first event at time one or the first event at time four leaves, respectively, three or four triples; both choices leave all three pairs and have the same unit deletion cost.

Reverse every arc and replace time t by $11 - t$, taking z as the source and assigning traversal time one. For either deletion, the same four vertices are reachable with earliest-arrival times $(0, 2, 4, 5)$ along the reversed path. The final reversed contact at time four comes from the undeleted original first event at time seven in both cases. Yet the original upper quota $W_3 \le 3$ accepts only the first deletion. No statistic consisting solely of these reached sets and earliest arrivals can distinguish the two choices, even after assigning fixed vertex weights.

This example separates the indicated summary from tuple multiplicities. It does not prove that reachability methods cannot be extended, nor that an auxiliary encoding is impossible.

C Reference implementation and local use

The implementation exports two functions:

```
solve_profiles(instance)
select_profile(result, accept, max_cost=None)
```

Its chronological state is exactly (q, ℓ, w); values are cost/mask pairs. Its branch aggregation uses exact min-plus convolution. Sparse dictionaries avoid materializing impossible states; no dominance rule across different count coordinates is used. Cost ties may choose any valid minimum-cost witness.

The JSON result distinguishes `original` and `baseline` counts, reports normalized `min_w2` and `R`, and lists profiles sorted by `(loss,w3)`. Each entry contains `w2`, `w3`, `optional_cost`, `total_cost`, and sorted `deleted_ids`. Invalid schemas or original graphs raise `ValueError`; a quota above the baseline returns an infeasible result. Unknown extra JSON fields are ignored; they do not become new features or constraints. The CLI permits arbitrary-length integer decimal encodings by temporarily disabling the interpreter's decimal-conversion limit when that setting exists. Importing the module does not change interpreter settings.

The implementation uses the Python standard library and supports Python 3.8 or newer. From the repository's `temporal-profiles/implementation` directory, run:

```
python3 demo.py
python3 profile_solver.py examples/nonmonotone.json --output result.json
```

The implementation archive documents how to run the exhaustive checker and reports the results. These examples run locally with the Python standard library.

D Reproducibility and evidence provenance

D.1 Canonical artifacts

The repository preserves the 5 November 2026 manuscript snapshot together with the compatibility-corrected implementation, examples, frozen corpus and validator, retained output files and summaries, the macOS validation summary, and the separate ten-event illustration check. Its frozen manifest identifies those artifacts. The present 12 November revision is distributed separately with its own source and checksums; it adds the deletion-propagation comparison while retaining the same algorithm, proofs of the existing results, and reported experimental evidence. The original supplement and historical audit records identify earlier sources and must not be confused with this manuscript revision.

The corrected implementation and frozen input identities are:

```
Solver:
a9f8f2c9f276055ab570940d784244192d177c16299f7a295ad75c481de7f4ba
Validator:
29ff8cfbe8df6f3aaedc9f7f0ae54c5a69a7081c805576a80d8f84760d5c8891
Corpus:
9a6979e08652fb949fa8eb7ca9d9a432204142c24e954d6fa046c172b631e67c
Protocol:
bcca092b7e763b8781fbdd261d3fff91f90c0dcdb70a965898f878de025e7297
```

The output digests recorded in the Mac summary and matched against the retained reference bytes are:

```
Candidate outputs:
faf9fc7f8960d46d81509ccbb711e83eb6c5a6657361b061113f7980131cad4a
```

```
Selection outputs:
133df11b935269866ea2766743a2b8037f0b88fd8b5be14bb2bf7743062e9f02
```

The macOS validation summary is identified by the following SHA-256 digest:

```
26eb54eaebed908e15fe45371c48f7b9703eb8838b30ad9900c67af82d7f574f
```

The Linux compatibility checks and the macOS validation run are documented separately in the implementation archive.

D.2 Aggregate transaction-cohort evidence

The separate Mac batch in Section 10.3 used the same solver hash as the frozen correctness suite above. Its aggregate results archive (`BATCH_RESULTS.zip`) has SHA-256

```
cd16912a10fe1f1e8444824f5a0648b57f2c2b05133e50be5bb1f24d522204c2
```

It contains the batch protocol, one result per query, one result per policy condition, and machine- and human-readable summaries. Each query has 18 policy conditions (three preservation levels, two target comparisons, and three budgets); repeated rounded lower quotas use the same table. The reported source, cohort, protocol and program hashes match the frozen configuration. The repository's frozen cohort file supplies the synthetic bank/account identifiers and cutoff times needed to select the same queries. The aggregate archive does not contain the original dataset, complete local contexts, full profiles, or native deletion-witness diagnostics; those detailed run files remained on the machine that ran the batch. The witness replays and exhaustive comparisons reported here were performed on that machine. Reaggregating the export checks outcome totals without independently replaying those witnesses. This batch and the worked application pilot are separate from the 375-history correctness suite.

D.3 The retained endpoint-specification correction

The initial checker specification omitted the requirement that a counted walk end at the root. For a three-event path, the checker therefore counted both length-two subwalks, although only one ends at the root. The solver already enforced this requirement and did not change. The corrected checker tests whether the final event ends at the root.

All 375 original histories and 1,779 original quota queries were preserved. The amendment added 84 missing thresholds derived from the corrected root baseline, yielding 1,863 queries. The archive retains the initial failure, both corpus/protocol/oracle snapshots, exact query additions, and the amended successful run. This correction changed the expected counts used for validation.

D.4 Build and run separation

The solver and the exhaustive checker run locally with the Python standard library. The manuscript builds with `latexmk -pdf main.tex` from this revision's `source` directory; the accompanying Mac build script places generated files in a separate build directory. Rebuilding the PDF does not rerun scientific tests. The implementation and validation archive and the Mac cohort runner are distinct artifacts from this manuscript-source revision.

Research workflow. Goutam Dyapa ran the reported macOS experiments. ChatGPT assisted with manuscript drafting and revision, code development, and validation tasks.

E Deletion-propagation encoding and a weighted formulation

This appendix derives a representation of the temporal problem in the GDP framework [13, Definition 4.1], and a weighted ILP extension. The construction concerns expressibility. It does not transfer the running-time guarantee of Theorem 2.1 to a general-purpose solver.

Use the normalization of Section 2. In particular, validate the original temporal graph before applying forced deletions, set $E_0 = E \setminus F$, and reserve the fixed cost $c(F)$. If $T_0 > N_2$, return infeasibility. Otherwise let $R = N_2 - T_0$. The original event identifiers are distinct even when endpoints and timestamps coincide.

E.1 Protected compatibility data and unit costs

Create a unary relation $\mathsf{Event}(i)$ with one row for every identifier in E_0. For $h \in \{2, 3\}$, let H_h contain exactly the ordered h-tuples of identifiers forming strictly time-increasing walks ending at the designated root in E_0. These relations have at most M^2 and M^3 rows. They can be constructed by polynomially many endpoint and binary-timestamp comparisons; the numeric lifetime is not expanded.

Protect each nonempty H_h by placing its identity query in the GDP preservation list with threshold $|H_h|$. Since source deletion cannot add rows to an identity view, no row of that relation can be deleted. An empty relation needs no protection requirement. The remaining view queries are ordinary conjunctive queries:

$$Q_2(i, j) :- H_2(i, j), \mathsf{Event}(i), \mathsf{Event}(j),$$
$$Q_3(i, j, k) :- H_3(i, j, k), \mathsf{Event}(i), \mathsf{Event}(j), \mathsf{Event}(k).$$

Strict comparisons and the root requirement are thus compiled into protected relation membership, without assuming built-in inequality syntax in GDP.

Proposition E.1 (Unit-cost profile encoding)**.** *Suppose all optional events have unit deletion cost. For every $0 \le \ell \le R$ and $0 \le w \le N_3$, a GDP instance of size polynomial in L has a feasible solution exactly when $C(\ell, w) < \infty$; its optimum is $C(\ell, w)$. An optimal source-deletion set yields an attaining original-event witness in polynomial time.*

Proof. Write $w_2 = N_2 - \ell$ and $w_3 = w$. For each $h \in \{2, 3\}$, put Q_h in the deletion list with requirement $N_h - w_h$ whenever this value is positive, and in the preservation list with requirement w_h whenever that value is positive. The former requires at most w_h surviving output rows; the latter requires at least w_h. Omit every zero requirement, consistent with the positive thresholds in GDP Definition 4.1. At the endpoints, the omitted inequality is automatic because $0 \le |Q_h| \le N_h$.

For each $e \in K$, preserve the Boolean singleton query $\mathsf{Keep}_e() :- \mathsf{Event}(e)$ with requirement one. Here e is the constant original identifier. Minimize deletions in the identity view $Q_E(i) :- \mathsf{Event}(i)$, with no maximization views. The protected compatibility rows cannot be deleted, and these singleton requirements prohibit deletion of events in K. Therefore every feasible source-deletion set is exactly a set $D_0 \subseteq O$ of optional event rows.

For any such D_0, a row of Q_h survives precisely when every event in its identifier tuple belongs to $E_0 \setminus D_0$. Its cardinality is therefore $W_h(E_0 \setminus D_0)$. All tuple identifiers are output coordinates, so set semantics does not merge parallel-event walks. A single deleted event affects all rows containing it in both views. Hence the two exact-count requirements hold exactly for the deletion sets defining $C(\ell, w)$, and the objective equals $|D_0|$. There are $O((M + 1)^3)$ data rows and $O(M + 1)$ view requirements, with polynomially encoded identifiers and thresholds. This proves the size, feasibility, optimum, and witness assertions. □

The full deletion witness is $F \cup D_0$ and its total cost is $c(F) + C(\ell, w)$. The GDP objective above concerns optional deletions only; forced cost remains the reserved constant.

E.2 Binary costs and full-profile recovery

For arbitrary nonnegative binary-encoded costs, use a binary retention variable x_e for every $e \in E_0$, fixed to one for $e \in K$. Introduce a binary variable z_q for each tuple $q \in H_2 \cup H_3$, and impose

$$z_q \le x_e \qquad (q \in H_2 \cup H_3,\ e \in q), \tag{21}$$

$$z_q \ge \sum_{e \in q} x_e - |q| + 1 \qquad (q \in H_2 \cup H_3), \tag{22}$$

$$\sum_{q \in H_2} z_q = N_2 - \ell, \qquad \sum_{q \in H_3} z_q = w. \tag{23}$$

The objective is

$$\min \left(c(F) + \sum_{e \in E_0} c_e (1 - x_e) \right). \tag{24}$$

For binary variables, (21)–(22) make z_q the conjunction of its event-retention decisions. Thus this ILP has optimum $c(F) + C(\ell, w)$ whenever the target is attainable, and is otherwise infeasible. A walk destroyed by several deletions still contributes only one lost tuple. There are $O((M+1)^3)$ variables and constraints, and all coefficients have polynomial bit length in L. In particular, retaining c_e as a binary coefficient avoids replication proportional to its value. This is an explicit weighted witness formulation, not a claim that arbitrary binary event costs are the literal objective of GDP Definition 4.1.

Solving each target independently and retaining an optimal witness recovers the full table. The number of target pairs is at most

$$(R+1)(N_3+1) \le (M^2+1)(M^3+1) = O((M+1)^5).$$

Consequently, full-profile output is expressible through polynomially many exact optimization calls. Their representation size does not bound their solving time in terms of R. The particular tractability results in Makhija and Gatterbauer [13, Section 6] do not automatically transfer under these joint exact-count requirements and cost coefficients; a proposed transfer must verify the relevant hypotheses and parameter bounds.

An elementary relaxation diagnostic. Let $b_1, b_2 : v \to u$ be distinct events at time zero, and let $c : u \to z$ be an event at time one. All three have unit cost and are optional. This depth-two instance has $(N_2, N_3) = (2, 0)$. With $T_0 = 1$ and target $(W_2, W_3) = (1, 0)$, the integral optimum is one: delete either middle event; deleting c misses the target. In the relaxation of the displayed ILP to $[0, 1]$, the assignment

$$x_{b_1} = x_{b_2} = 1, \qquad x_c = \tfrac{1}{2}, \qquad z_{(b_1,c)} = z_{(b_2,c)} = \tfrac{1}{2}$$

is feasible and costs $\frac{1}{2}$. Thus this relaxation is not automatically integral on the temporal class. This diagnostic concerns the displayed formulation; it excludes neither stronger formulations nor other algorithms. It does not replace the structural proof of Theorem 2.1 or prove that no earlier result can imply its guarantee.

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